



Finding $A:B:C$ where
 $mA = nB = pC$

Que: If $3A = 5B = 6C$, find $A : B : C$?



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Que: If $3A = 5B = 6C$, find $A : B : C$?

$$3A = 5B = 6C = K$$

$$3A = K$$

$$A = \frac{K}{3}$$

$$5B = K$$

$$B = \frac{K}{5}$$

$$6C = K$$

$$C = \frac{K}{6}$$

$$\text{LCM}(3, 5, 6) \\ \Rightarrow 30$$

$$\underline{\underline{A : B : C = 10 : 6 : 5 \checkmark}}$$

$$A : B : C = \frac{K}{3} : \frac{K}{5} : \frac{K}{6}$$

$$A : B : C = \frac{1}{3} : \frac{1}{5} : \frac{1}{6} \checkmark$$

$$= \frac{10}{30} : \frac{6}{30} : \frac{5}{30}$$

Que: If $3A = 5B = 6C$, find $A : B : C$?

$$mA = nB = pC$$

↑ ↑

$$3A = 5B = 6C$$

$$\underline{A : B : C} = \underline{\frac{1}{m} : \frac{1}{n} : \frac{1}{p}}$$

$$A : B : C = \frac{1}{3} : \frac{1}{5} : \frac{1}{6} \checkmark$$

Que: If $2A = 3B = 4C$, find $A : B : C$?

Que: If $2A = 3B = 4C$, find $A : B : C$? $\leftarrow$

$$2A = 3B = 4C$$

$$A : B : C = \frac{1}{2} : \frac{1}{3} : \frac{1}{4}$$

$\times 6 \quad \times 4 \quad \times 3$

$$6 : 4 : 3 \checkmark$$

$$\text{LCM}(2, 3, 4) = 12$$

$$\frac{6}{12} : \frac{4}{12} : \frac{3}{12}$$

$$A : B : C = 6 : 4 : 3 \checkmark$$